

CYBER SECURITY PROJECT 4

System Vulnerability Checklist & Security Assessment Using Kali Linux

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Introduction

Cybersecurity plays an important role in protecting computer systems, networks, and sensitive information from cyber threats and unauthorized access. This project focuses on performing a practical system vulnerability assessment using Kali Linux tools and Linux terminal commands.

Tool / Command	Purpose
hostnamectl	Display system information
sudo ufw status	Check firewall status
nmap localhost	Scan localhost for open ports
sudo ss -tulnp	Display open ports and services
apt list --upgradable head -20	Check outdated packages

1. System Information Analysis

```
(raja@raja)-[~]
$ hostnamectl
  Static hostname: raja
           Icon name: computer-laptop
           Chassis: laptop
  Asset Tag: 5CG423D5W3
  Machine ID: 4d268fdd95f54880bfc8aaeff50f7948
  Boot ID: 4c2116babd8f45ecbfd9bf3938c96cd4
Operating System: Kali GNU/Linux Rolling
           Kernel: Linux 6.19.11+kali-amd64
           Architecture: x86-64
  Hardware Vendor: Hewlett-Packard
  Hardware Model: HP ProBook 645 G1
  Hardware SKU: D2Z92AV
Hardware Version: A3009DD10303
Firmware Version: L72 Ver. 01.35
  Firmware Date: Tue 2015-10-20
  Firmware Age: 10y 7month 2w

(raja@raja)-[~]
$
```

2. Firewall Configuration Analysis

```
raja@raja: ~
(raja@raja)-[~]
$ sudo ufw status
Status: inactive

(raja@raja)-[~]
$
```

3. Localhost Vulnerability Scan

```
(raja@raja)-[~]
$ nmap localhost
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-03 21:03 +0500
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000090s latency).
Other addresses for localhost (not scanned): ::1
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh

Nmap done: 1 IP address (1 host up) scanned in 0.11 seconds

(raja@raja)-[~]
$
```

4. Open Ports & Running Services Analysis

```
These make your project look advanced:
(raja@raja)-[~]
$ sudo ss -tulnp
ss: netstat failed: Invalid argument (95)
(raja@raja)-[~]
$ ss -tulnp
State      Recv-Q Send-Q Local Address:Port Peer Address:Port
tcp        0      0 0.0.0.0:22          0.0.0.0:0*      users:((("sshd",pid=753,fd=6))
tcp        0      0 127.0.0.1:35717    0.0.0.0:0*      users:((("code",pid=106110,fd=36))
tcp        0      0 :::22            ::::0*          users:((("sshd",pid=753,fd=7))
tcp        0      0 :::1716          ::::0*          users:((("kdeconnectd",pid=2040,fd=27))
tcp        0      0 [fe80::3c38:84fc:bf3e:f0f5]:::546 [:::]:546      users:((("NetworkManager",pid=583,fd=31))
tcp        0      0 *:*              *:*:0*          users:((("kdeconnectd",pid=2040,fd=26))
tcp        0      0 *:*              *:*:0*          users:((("brave",pid=88427,fd=21))
tcp        0      0 0.0.0.0:22        0.0.0.0:0*      users:((("sshd",pid=753,fd=6))
tcp        0      0 127.0.0.1:35717    0.0.0.0:0*      users:((("code",pid=106110,fd=36))
tcp        0      0 :::22            ::::0*          users:((("sshd",pid=753,fd=7))
tcp        0      0 *:*              *:*:0*          users:((("kdeconnectd",pid=2040,fd=27))

(raja@raja)-[~]
$
```

5. Software Update & Patch Management

```
—(raja@raja)-[~]  
—$ apt list --upgradable | head -20  
  
WARNING: apt does not have a stable CLI interface. Use with caution in scripts.  
  
Listing...  
zip/kali-rolling 26.01+dfsg-2 amd64 [upgradable from: 26.00+dfsg1-2]  
dduser/kali-rolling 3.155 all [upgradable from: 3.154]  
dw-gtk3-kali/kali-rolling 2026.2.0 all [upgradable from: 2025.2.0]  
ircrack-ng/kali-rolling 1:1.7+git20230807.4bf83f1a-3 amd64 [upgradable from: 1:  
.7+git20230807.4bf83f1a-2+b1]  
mass/kali-rolling 5.1.1-0kali1 amd64 [upgradable from: 5.0.1-0kali4]  
nt-optional/kali-rolling 1.10.17-1 all [upgradable from: 1.10.15-1]  
nt/kali-rolling 1.10.17-1 all [upgradable from: 1.10.15-1]  
pache2-bin/kali-rolling 2.4.67-1 amd64 [upgradable from: 2.4.66-8]  
pache2-data/kali-rolling 2.4.67-1 all [upgradable from: 2.4.66-8]  
pache2-utils/kali-rolling 2.4.67-1 amd64 [upgradable from: 2.4.66-8]  
pache2/kali-rolling 2.4.67-1 amd64 [upgradable from: 2.4.66-8]  
pparmor/kali-rolling 4.1.7-2 amd64 [upgradable from: 4.1.6-3]  
rk/kali-rolling 4:26.04.0-1 amd64 [upgradable from: 4:25.12.3-1]  
t-spi2-common/kali-rolling 2.60.3-1 all [upgradable from: 2.60.0-1]  
t-spi2-core/kali-rolling 2.60.3-1 amd64 [upgradable from: 2.60.0-1+b1]  
tril-common/kali-rolling 1.28.4-1 all [upgradable from: 1.28.2-1]  
tril/kali-rolling 1.28.4-1 amd64 [upgradable from: 1.28.2-1+b1]  
aloo6/kali-rolling 6.23.0-2 amd64 [upgradable from: 6.23.0-1]  
ase-files/kali-rolling 1:2026.1.2 amd64 [upgradable from: 1:2026.1.1]
```

Conclusion

This project provided practical experience in conducting a vulnerability assessment using Kali Linux tools and Linux terminal commands. The assessment improved understanding of practical cybersecurity analysis and vulnerability management techniques.